Dachi Tchotashvili

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EDUCATION

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI) **Abu Dhabi, UAE**

BSc in AI Engineering | GPA: 3.69 / 4.00 Aug 2025 – Present | Expected graduation: 2029

- Robotics Club Treasurer & Event Organizer; recipient of a merit-based HH Sheikh Tahnoon Bin Zayed scholarship.
- **Coursework**: Deep Learning, Probability & Statistics, Multivariable Calculus, Data Engineering & Science.

SELECTED AWARDS

- **International Physics Olympiad:** 2 silver medals (2022, 2023) and 1 bronze medal (2021).
- **World Robot Olympiad:** national selection overall winner and international bronze medal (2023).
- **National Sumo Robot Championship:** 3rd place, Robo Battle (GRA.ge, 2024).
- **Georgian Young Physicists' Tournament:** ranked 1st nationally (2023).

ENGINEERING EXPERIENCE & PROJECTS

6-axis Robot Arm Platform | Undergraduate Researcher at MBZUAI **Abu Dhabi, UAE** (Aug 2026 – Present)

- Designed a 6-axis arm with automatic tool changer for vision-guided automation using AprilTags; 3D printed & assembled a prototype.
- Modeled joint motion in **Isaac Sim** and refined internal cable routing to preserve joint travel and AprilTag visibility.
- Developing experiments to collect joint position-versus-time data, model **stochastic error tracking**, and evaluate motor-target compensation for improved trajectory tracking; working under **Prof. Jorge A. Amador Herrera's** supervision.

Active Spoiler Prototype for BMW 6 Series (F06) | Designer **Tbilisi, Georgia** (Jun 2026 – Present)

- Built body-integrated spoiler pop-up wings with independent left/right actuation via 4 electric motors and four-bar linkages.
- Implemented dual-ESP32 control with 1 kHz servo position loops & 14-bit encoder feedback (0.022°/count).
- Mapped 100 Hz MPU6050 accelerometer readings to independent wing-angle targets using custom tuning software.
- Simulated in **Ansys Discovery**, predicted 300 ± 20 N combined aerodynamic braking force at 150 km/h and 60° deployment.

Sherekilebi Labs | Founder **Tbilisi, Georgia** (Jun 2026 – Present)

- Established an **independent robotics lab** with 4 peers in 2 months, equipping it with $2,500 in prototyping tools.
- Collaborated on 6 projects using CAD, FDM 3D printing, ESP32/C++ firmware, sensor integration, and motor control.
- Repurposed an Xbox 360 Kinect via custom low-level drivers (**libfreenect**) into a unified **RGB-D** and **4-mic acoustic array** sensor suite, fusing real-time 3D spatial tracking and directional audio on an 8GB RTX 3070 for sub-500ms voice/gesture automations.

AI Smart Hydroponic Greenhouse | Team Member **Tbilisi, Georgia** (Mar 2023 – May 2023)

- Built a 2 × 1 × 1 m greenhouse in a team of 5; automated climate, lighting, watering, and nutrient dosing with ESP32 & sensors.
- Implemented a **PyTorch EfficientNet-B0** pipeline with 224 × 224 grayscale preprocessing and a two-class output layer to classify lettuce leaves as healthy or diseased; won the **National STEM Competition.**

Independent Physics Research | Student Researcher **Tbilisi, Georgia** (Oct 2019 – Jul 2023)

- Investigated 8 open physics problems with Free University professors for the **International Young Physicists' Tournament** (IYPT).
- Used simplified **Navier–Stokes equations** to model 3 flow regimes and transitions for the Unsinkable Disk (IYPT 2022 problem).
- Simulated **Euler-Bernoulli** beam model, magnetic forces, and coupled oscillations in **MATLAB**; Defended solutions at IYPT 2023.

WORK EXPERIENCE & LEADERSHIP

European Physics Olympiad | Academic Committee Member **Kutaisi, Georgia** (Jan 2024 – Jul 2024)

- Co-developed experimental exam on **piezoelectricity** with 15 international committee members, mostly physics professors.
- Designed apparatus and contributed to calibration & data analysis; helped hand-assemble 280 standardized experimental kits.

Online STEM School | School Administrator **Tbilisi, Georgia** (Sep 2023 – Jun 2024)

- Co-developed a 2-year STEM curriculum with 10 instructors, integrating IoT hardware and Tinkercad simulations.
- Managed Google Classroom and Excel records for 400+ students; resolved classroom access, IoT setup, and parent inquiries.

COMMUNITY INVOLVEMENT (VOLUNTEERING)

International Young Naturalists' Tournament | Volunteer **Tbilisi/Batumi, Georgia** (2022, 2024, 2026)

- Guided debutant Team Uganda through all events (2022); coordinated logistics for 10+ international delegations (2024, 2026).

European Girls' Mathematical Olympiad | Grader & Moderator **Tskaltubo, Georgia** (Apr 2024)

- Contributed to fair grading of 212 contestants with a 30-member committee; discussed grading decisions with 30+ contestants.

SKILLS & LANGUAGES

Engineering: Ansys Discovery, Fusion 360, Isaac Sim, KiCad, Arduino IDE.
Programming & ML: Python, C++, MATLAB, PyTorch, scikit-learn, pandas, SQL.
Languages: English (advanced), Georgian (native), Russian (B1).